
Modelling of a PMUT Pulse-Echo in a Hydraulic Cylinder

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Abstract

Piezoelectric micromachined ultrasonic transducers (PMUTs) are usually designed for short-range imaging in soft tissue near 1 MHz. I ask a different question: can a PMUT range the piston face of a hydraulic cylinder by pulse-echo down a 3 m oil-filled steel bore, and what actually limits it? Starting from the published system-level transducer model of Dangi and Pratap (2017), which radiates into air over sub-half-metre distances and carries no noise model, I add the missing pieces: an acoustic-channel model for the tube treated as a waveguide (transfer-matrix wall coupling, confinement instead of spherical spreading, frequency dependent oil absorption, and a near-total far-end reflector), a first-principles charge-amplifier receiver-noise model, and a two-dimensional finite-difference wave simulation for visualization. Three results emerge that overturn the default intuition. First, the conventional choice of frequency is backwards: oil absorption scales as f^2 over the fixed round trip, so the echo amplitude collapses as $\exp(-kf^2)$, giving 1920 dB of round-trip loss at 1 MHz versus 120 dB at 250 kHz; the usable band lies near and below 135 kHz, not at 1 MHz. Second, when transmit and receive are forced to derive from one electromechanical coupling (so the two-way link is passive, which the published sensitivities are not: they imply voltage gain above unity), the detection boundary sits at roughly 135 kHz for a representative 20 dB/m oil. Third, the binding constraint is the receiver, not the channel: a thin-film-PZT clamped capacitance of order 479 nF feeding a low-noise charge amplifier sets a noise floor tens of dB above the convenient “10 μ V” placeholder. I give the link budget, the reciprocity relation, the noise model, a 100/135/200 kHz comparison spanning detectable to buried, and matching wave-propagation movies. Absolute signal-to-noise carries order-of-magnitude uncertainty because the source paper’s own transmit and receive sensitivities disagree by more than 100 $\times$; the relative structure, the cliff, and the receiver-limited regime are robust.

1 Introduction

The position of the piston inside a hydraulic cylinder, together with the pressure of the oil that drives it, fixes the instantaneous position and force state of the actuator, which is what closed-loop motion control, condition monitoring, and safety interlocks need. Measuring both from a single device at the cylinder end cap, without breaching the bore or adding an external rod-mounted scale, would be valuable. This work asks whether a piezoelectric micromachined ultrasonic transducer (PMUT) integrated into the end cap can do it by pulse-echo: ranging the steel piston face along the oil-filled bore to recover position, while the same acoustic path and the pressure dependence of the oil properties inform the force estimate. PMUTs are small, low-power, and CMOS-compatible [Jung et al., 2017, Przybyla et al., 2011], which suits embedding in an end cap. Almost all PMUT design practice, however, targets short-range medical and fingerprint imaging, where the operating frequency is pushed toward 1 MHz and above for axial resolution and the medium is soft tissue or water over millimetres to a few centimetres.

A 100 kHz pulse launched at the end cap travels the 3 m oil column, reflects off the piston face, and returns

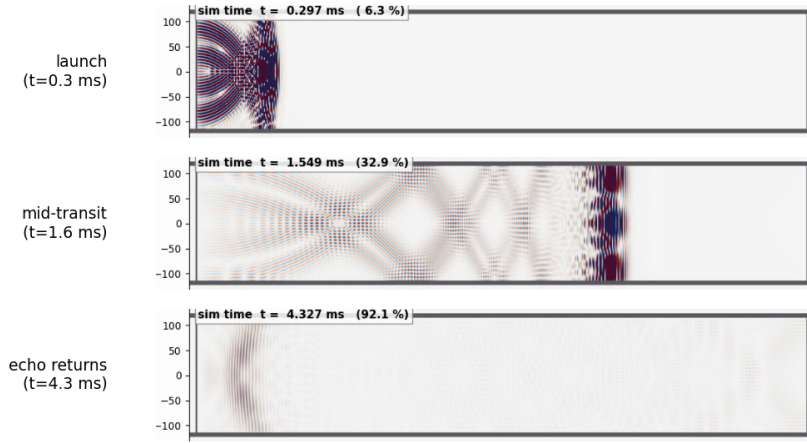


Figure 1: The central phenomenon. A 100 kHz acoustic pulse launched by the PMUT at the cylinder end cap travels the 3 m oil column (launch, top), spreads and fills the bore (mid-transit, middle), reflects off the steel piston face, and returns to the transducer about 4.2 ms later (echo returns, bottom), where its time of flight gives the piston position. Red and blue are positive and negative pressure; grey is steel. Frames are from the two-dimensional FDTD simulation of Section 4. At 200 kHz and above the pulse is absorbed before it can complete the round trip (Figure 4); that frequency dependence is the main result of this paper.

The cylinder is a very different regime: a transducer at one end of an oil-filled steel bore up to 3 m long, ranging the piston face. I model the bore as a 3 m tube (outer diameter 300 mm, wall 10 mm) with the piston represented by a thick-steel specular reflector at the far end, the worst case of full extension. The starting point for the transducer is the system-level analytical model of Dangi and Pratap [2017], which compactly relates a multilayer PMUT’s transmit pressure, receive charge, and quality factor to its geometry through three nearly invariant mode-shape factors. That model is validated for radiation into air over distances below 0.5 m and contains no treatment of layered solid/liquid channels, of propagation to metres, of pulse-echo, or of noise. It therefore cannot, by itself, answer the ranging question. Figure 1 shows the phenomenon at the heart of the study: the pulse making its round trip down the bore.

I extend it into an end-to-end link model and report three findings. (1) The operating frequency is set by a steep absorption cliff: because the oil attenuation coefficient grows as f^2 , the round-trip loss over 6 m is 11 dB at 75 kHz, 120 dB at 250 kHz, and 1920 dB at 1 MHz; the intuitive high-frequency choice is not merely suboptimal but fatal, and the usable band is far lower. (2) Confinement, not radiation, governs feasibility: the tube is a waveguide, so the spherical spreading that dominates a free-field budget is replaced by a bounded cross-section fill loss. (3) When the transmit and receive paths are required to come from a single coupling so the link is passive, and noise is modeled from the transducer capacitance and amplifier rather than assumed, the detection boundary lands near 135 kHz and is set by the receiver electronics rather than by the acoustic channel.

My contribution is a careful, reciprocity-consistent system model and the resulting design inferences, not a new transducer or a new experiment. I am explicit about a calibration caveat: the source paper’s published transmit and receive sensitivities are mutually inconsistent by more than two orders of magnitude, so absolute signal-to-noise ratios (SNRs) are uncertain to within roughly that factor. The robust outputs are the relative trends, the f^2 cliff, the passivity constraint, and the receiver-limited regime, all of which are independent of the absolute calibration. I also show that the low-frequency band and the cliff forces is where single-reflector carrier-phase ranging meets the application’s sub-3 mm (0.1 % of 3 m) precision target.

2 The source transducer model

A PMUT is a clamped circular multilayer plate with a piezoelectric layer (here lead zirconate titanate, PZT) between electrodes on a passive structural layer. Applying a voltage produces an unbalanced in-plane stress and hence a bending moment, so the plate vibrates flexurally and pumps the adjacent fluid; conversely an incident pressure flexes the plate and generates charge. Dangi and Pratap [2017], building on the clamped-plate analysis of Sammoura et al. [2012], reduce this to a system-level electro-mechano-acoustic model. I restate that model here, faithfully but compactly, so the present work is self-contained and so later equations refer to this paper's own numbering rather than to equation numbers in the source.

Assumptions. The source model rests on (i) classical (Kirchhoff) thin-plate theory for a clamped circular plate carrying an in-plane pre-tension; (ii) a top electrode treated as covering the whole suspended plate, valid when the electrode is thin compared with the stack, so the multilayer behaves as one equivalent plate; and (iii) neglect of the initial curvature that arises from non-uniform radial residual stress, so the only forcing is the piezoelectric bending moment and the deflection is measured relative to the static equilibrium at the operating residual tension. I adopt these same assumptions.

Structural response. The clamped plate with pre-tension obeys

$$D_e \nabla^4 w - T_e \nabla^2 w + \rho_h \frac{\partial^2 w}{\partial t^2} = f(r, \theta, t), \quad (1)$$

with mass per area $\rho_h = \sum_i \rho_i h_i$, net in-plane tension $T_e = \sum_i \sigma_i h_i$, and equivalent flexural rigidity

$$D_e = \sum_i \frac{E_i}{1 - \nu_i^2} \left(\frac{h_i^3}{12} + z_i^2 h_i \right), \quad (2)$$

summed over layers i of Young's modulus E_i , Poisson ratio ν_i , thickness h_i , and mid-plane offset z_i from the neutral plane. The non-dimensional ratio of tension to rigidity,

$$\kappa^2 = T_e a^2 / D_e \quad (a = \text{plate radius}), \quad (3)$$

sets whether the device behaves as a plate ($\kappa^2 \ll \alpha_{00}^2$) or a tensioned membrane ($\kappa^2 \gg \alpha_{00}^2$). The clamped eigenproblem gives the first natural frequency

$$\omega_{00} = \alpha_{00} \beta_{00} \sqrt{\frac{D_e}{\rho_h a^4}} = \sqrt{\frac{\alpha_{00}^2 D_e}{\rho_h a^4} (\alpha_{00}^2 + \kappa^2)}, \quad \beta_{00}^2 = \alpha_{00}^2 + \kappa^2, \quad (4)$$

where α_{00} is the first clamped eigenvalue, which the source finds runs only from 3.19 (plate) to 2.14 (membrane). The two limits are $\omega_{00} \approx (\alpha_{00}^2/a^2) \sqrt{D_e/\rho_h}$ (plate, scaling as $1/a^2$) and $\omega_{00} \approx (\alpha_{00}/a) \sqrt{T_e/\rho_h}$ (membrane, scaling as $1/a$).

Mode-shape factors. With the normalized first mode $\psi_{00}(r)$, three participation factors carry the geometry into the lumped model,

$$\Lambda' = r_p \left. \frac{\partial \psi_{00}}{\partial r} \right|_{r=r_p}, \quad \Lambda_1 = \int_0^a \frac{2\psi_{00} r}{a^2} dr, \quad \Lambda_2 = \int_0^a \frac{2\psi_{00}^2 r}{a^2} dr, \quad (5)$$

representing the piezoelectric-moment, uniform-pressure, and inertial participations, with r_p the top-electrode radius. The source's central structural finding is that these are nearly invariant over six decades of κ^2 ($\Lambda' \approx 0.99$ – 1.24 , $\Lambda_1 \approx 0.33$ – 0.42 , $\Lambda_2 \approx 0.20$ – 0.26), so they may be treated as constants; I use the plate-regime values.

Forced response. The piezoelectric drive is a moment load per unit voltage $M_p = e_{31,f} z_p$, with $e_{31,f}$ the effective transverse piezoelectric coefficient and z_p the PZT mid-plane offset from the neutral plane. Near the first resonance the centre deflection per unit drive voltage and per unit incident pressure are

$$\frac{w_0}{V_{\text{in}}} = \frac{2M_p}{\rho_h a^2 \omega_{00}^2} \frac{\Lambda'}{\Lambda_2} \mathcal{D}(\gamma), \quad \frac{w_0}{P_{\text{in}}} = \frac{1}{\rho_h \omega_{00}^2} \frac{\Lambda_1}{\Lambda_2} \mathcal{D}(\gamma), \quad (6)$$

with $\gamma = \omega/\omega_{00}$ and the second-order resonance factor $\mathcal{D}(\gamma) = [(1 - \gamma^2)^2 + (2\zeta\gamma)^2]^{-1/2}$ (ζ the damping ratio).

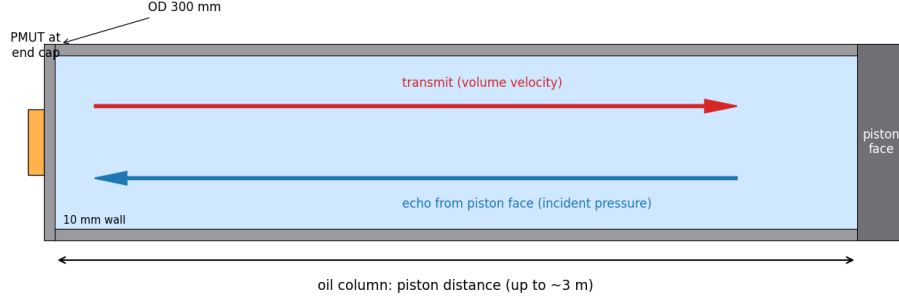


Figure 2: Pulse-echo geometry. A PMUT array at the near end launches into the oil column through the 10 mm steel end-cap; the wave travels 3 m, reflects off the thick-steel far end, and returns to the same transducer. The tube walls confine the beam, so the channel is a waveguide rather than free space.

Lumped circuit and transfer functions. The device maps to an electrical port (clamped capacitance C_s) coupled through a transformer of ratio ϕ to a mechano-acoustic branch (compliance C_m , mass L_m , loss R_m) loaded by the piston radiation impedance D_{ac} . The coupling and branch elements are

$$\phi = \frac{2e_{31,f}z_p}{a^2} \frac{\Lambda_1'}{\Lambda_1}, \quad C_m = \frac{\pi a^2}{\rho_h \omega_{00}^2} \frac{\Lambda_1^2}{\Lambda_2}, \quad L_m = \frac{\rho_h}{\pi a^2} \frac{\Lambda_2}{\Lambda_1^2}, \quad (7)$$

with $\omega_{00}^2 = 1/(L_m C_m)$ and the resonant mechanical impedance $Z_m = R_m + \text{Re}(D_{ac}) = \omega_{00} L_m / Q$ set by the quality factor Q . The transmit volume velocity and the receive charge sensitivity follow from the *same* ϕ and Z_m ,

$$\frac{\dot{q}}{V_{in}} = \frac{\phi}{Z_m}, \quad \left| \frac{Q_{rx}}{P_{in}} \right| = \frac{2\pi Q}{\omega_{00}^2} \frac{e_{31,f}z_p}{\rho_h} \frac{\Lambda_1' \Lambda_1}{\Lambda_2}, \quad (8)$$

and the radiated far-field pressure is $P_{tx} = j\rho_{ac}\omega\dot{q}/(2\pi R)H(\Theta)e^{-\alpha R}$, with $H(\Theta)$ the piston directivity. Equation (8) is the receive charge sensitivity that the present work later re-derives by reciprocity (Section 3.2).

Source results and validation. The source reports that, for a fixed layer stack, the transmitted pressure is nearly independent of device size and tension; the receive sensitivity scales as $1/\omega_{00}^2$; plate-regime deflection scales as a^2 and membrane deflection as a^0 ; and the optimal electrode ratio r_p/a runs from about 0.65 (plate) to 0.95 (membrane). It validates against fabricated plate-type (10 μm Si, $\kappa^2 \approx 0$) and membrane-type (300 nm SiO₂, $T_e \approx 130 \text{ N/m}$, $\kappa^2 \approx 1000$) PMUTs, with measured first-mode resonances matching the model and a measured plate-type deflection sensitivity of about 3 nm/V at 126 kHz for a 1 mm device, which I use later as the calibration anchor. Crucially for the present work, this model is validated only for radiation into *air* over sub-half-metre distances and contains no layered solid/liquid channel, no propagation to metres, no pulse-echo, and no noise. Equations (1)–(8) are what I reuse for the transducer; everything outside the device is built in Section 3.

3 System model

Figure 2 shows the geometry. A PMUT (or, more realistically, a small PMUT array forming an effective aperture of radius a_{ap}) sits at the near end of a steel tube of outer diameter 300 mm and wall thickness 10 mm, so the inner oil radius is $R_{in} = 140 \text{ mm}$. The tube is filled with a synthetic mineral oil and is $L = 3 \text{ m}$ long. The far end is a thick steel block, thick enough that the steel/air interface behind it is irrelevant, so it acts as a near-total acoustic reflector. The transducer both transmits and, after the round trip, receives.

3.1 Acoustic channel as a link budget

I track acoustic intensity through the round trip as a product of factors, then convert to an incident pressure at the receiver. The transmitted power crosses the near steel cap into the oil, propagates and attenuates to the far end, reflects, returns, and crosses the cap again. Writing each stage as an intensity factor,

$$I_{\text{back}} = I_{\text{face}} \cdot \underbrace{\tau_{\text{wall}}}_{\text{out}} \cdot \underbrace{\left(\frac{a_{\text{ap}}}{R_{\text{in}}}\right)^2}_{\text{fill}} \cdot \underbrace{\Gamma^2}_{\text{far end}} \cdot \underbrace{e^{-2\alpha(f)(2L)}}_{\text{oil absorption}} \cdot \underbrace{\tau_{\text{wall}}}_{\text{return}}. \quad (9)$$

Three of these terms carry the physics. The wall term τ_{wall} is the intensity transmission of the 10 mm steel layer, computed from the standard single-layer acoustic transfer matrix between the transducer-side medium and the oil; for a well-coupled (bonded) transducer it is a flat -10 dB per crossing, while for a dry, air-backed transducer it develops deep resonance windows at multiples of $c_{\text{steel}}/2d$. The confinement term replaces free-field spherical spreading: in an unbounded fluid the round-trip on-axis pressure falls as the square of $a^2/(\lambda z)$, costing about -41 dB at 3 m, but in the tube the energy that fills the cross-section is conserved and the only geometric loss is the fraction $(a_{\text{ap}}/R_{\text{in}})^2$ that the receive aperture re-samples, here -28.9 dB, and falling to 0 dB as the aperture fills the tube. The absorption term uses the classical thermoviscous law [Kinsler et al., 2000] in which the coefficient grows quadratically with frequency,

$$\alpha_{\text{dB}}(f) = \alpha_0 (f/f_{\text{ref}})^2, \quad \alpha_0 = 20 \text{ dB/m at } f_{\text{ref}} = 250 \text{ kHz}. \quad (10)$$

This single term dominates the frequency dependence of the whole system and is the source of the result in Section 5.1. A small additional waveguide wall-bounce loss of 0.5 dB/m is included. The incident pressure at the receiver is $P_{\text{inc}} = \sqrt{2Z_{\text{steel}}I_{\text{back}}}$.

3.2 Reciprocity-consistent transmit and receive

A passive, reciprocal transducer cannot present a two-way (transmit then receive) voltage transfer greater than unity. Anchoring the two paths independently, transmit from an assumed launched power and receive from the charge-sensitivity formula of Eq. (8), does not respect this: it gives an end-to-end transfer above unity at low frequency (a 140 V echo from a 10 V drive), which is unphysical. The correct construction makes both paths follow from a single electromechanical coupling, so the receive sensitivity is fixed by the transmit response rather than chosen independently. In the lumped model the transmit volume-velocity response and the receive charge sensitivity share the transformer ratio ϕ and the mechanical impedance Z_m , which forces the reciprocity relation

$$\left. \frac{\dot{q}}{V_{\text{in}}} \right|_{\text{transmit}} = \omega_{00} \left. \frac{Q}{P_{\text{in}}} \right|_{\text{receive}}, \quad (11)$$

where $\dot{q}$ is the volume velocity, ω_{00} the resonance, and Q/P_{in} the charge per unit incident pressure. I anchor the absolute scale to the paper's *measured* transmit (a 1 mm, 126 kHz device deflects about 3 nm/V at resonance, their Fig. 9) and *derive* the receive sensitivity from Eq. (11), rather than using the charge-sensitivity formula of Eq. (8) in isolation. The aperture is treated as an array of elements each sized to resonate at the operating frequency; the element count cancels in the SNR, so the result is aperture self-consistent. With this construction the implied transmit is physically sane (55 mW launched, face velocity 5.5 mm/s, face pressure 254 kPa), and the two-way transfer never exceeds $V_{\text{echo}}/V_{\text{in}} = 0.77$ across the band, restoring passivity. The reciprocity-derived receive sensitivity at 250 kHz is 27 fC/Pa, about $41\times$ smaller than the value 1130 fC/Pa from the charge-sensitivity formula Eq. (8); the two numbers are not mutually consistent, which is the origin of the calibration caveat stated throughout.

3.3 Receiver noise

The received charge is read by a charge amplifier with feedback capacitance C_f . The dominant noise is the amplifier input voltage noise e_n multiplied by the capacitive noise gain (C_{in}/C_f) , where $C_{\text{in}} = C_s + C_{\text{stray}} + C_f$ and C_s is the PMUT clamped capacitance. I add the amplifier current noise through the feedback transimpedance and the PZT dielectric-loss (Johnson) noise, and integrate over the resonant equivalent noise bandwidth $\text{ENBW} = (\pi/2)(f_0/Q)$:

$$V_{\text{noise}} = \sqrt{(e_n C_{\text{in}}/C_f)^2 + (e_i Z_f)^2 + 4k_B T G Z_f^2} \sqrt{\text{ENBW}}. \quad (12)$$

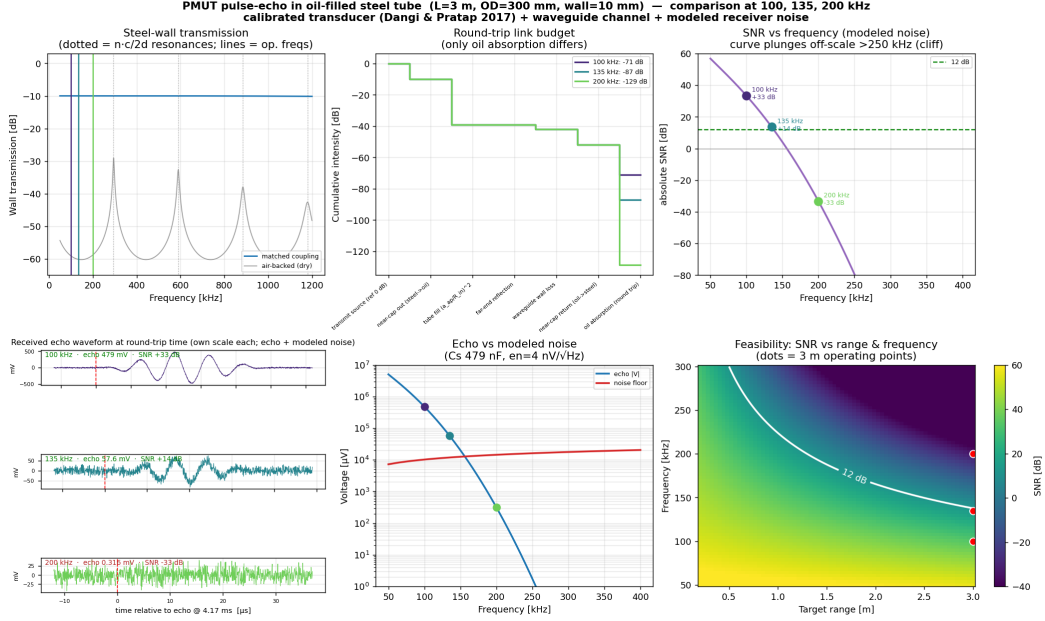


Figure 3: Calibrated comparison at 100/135/200 kHz. (1) Steel-wall transmission: flat for matched coupling, resonant if dry. (2) Round-trip link budget; only the oil-absorption term differs between frequencies. (3) Absolute SNR versus frequency with the three operating points marked; the curve plunges off-scale above 250 kHz. (4) The actual received echo waveform plus modeled noise at each frequency, each on its own absolute scale: a clean burst at 100 kHz, an echo emerging from noise at 135 kHz, and pure noise at 200 kHz. (5) Echo voltage and the modeled noise floor versus frequency, crossing near 135 kHz. (6) Feasibility map of SNR over range and frequency, with the 12 dB contour and the 3 m operating points.

Thin-film PZT (650 nm thick) over a 10 mm aperture gives a large clamped capacitance of order 479 nF, so with a low-noise $e_n = 4 \text{ nV}/\sqrt{\text{Hz}}$ the voltage-noise term dominates and the noise floor is of order tens of millivolts referred to the amplifier output, far above a convenient $10 \mu\text{V}$ placeholder. The link is therefore receiver-limited.

4 FDTD visualization

To see the propagation I built a two-dimensional finite-difference time-domain solver on an axial slice of the tube, following the explicit grid approach of WaveSimulator2D [0x23 (Michael Schardt), 2022] but using a variable-density velocity-pressure staggered scheme [Virieux, 1986] so that reflections follow the true acoustic impedance $Z = \rho c$. The oil sound speed (1440 m/s) is exact, so the round-trip time-of-flight of about 4.17 ms is physical. To keep the time step affordable, the steel sound speed is reduced while its density is kept high, which preserves the impedance contrast and hence the strong reflections; this is a visualization trade-off and the SNR numbers come from the link budget of Section 3, not from the FDTD. Each rendered movie has three synchronized panels: the wave field, the transmitter drive, and the received signal with the modeled noise, plus a running readout of physical simulation time.

5 Results

Figure 3 compares the calibrated model at 100, 135, and 200 kHz, the frequencies that bracket the detection boundary.

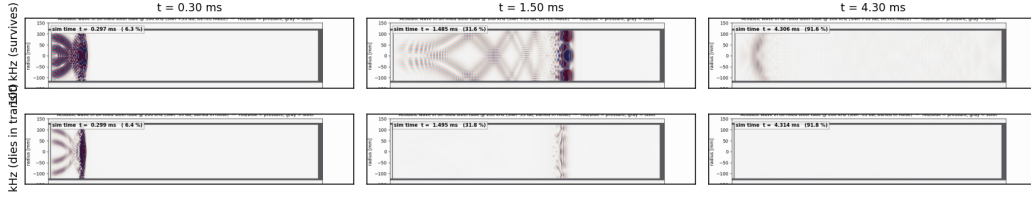


Figure 4: FDTD acoustic field on an axial slice. Top: at 100 kHz the pulse travels the full 3 m, reflects, and returns (echo present near the near end at 4.30 ms). Bottom: at 200 kHz the pulse is absorbed before completing the round trip, so no echo returns. The contrast is the f^2 absorption cliff made visible. The full three-panel movies for each frequency are released at <https://github.com/rayjyotir5/pmut-tube-echo/tree/main/media> (files 100khz.mp4, 135khz.mp4, 200khz.mp4).

5.1 The f^2 absorption cliff sets the frequency

Because the absorption coefficient in Eq. (10) grows as f^2 , the dB loss over the fixed 6 m round trip also grows as f^2 and the echo amplitude collapses as $\exp(-kf^2)$. The round-trip oil loss is 11 dB at 75 kHz, 120 dB at 250 kHz, and 1920 dB at 1 MHz. The receive sensitivity grows only as $1/f^2$ and the noise as $\sqrt{f}$, so neither offsets the collapse. The practical consequence is stark and runs against ultrasonic-imaging intuition: a frequency chosen for axial resolution is exactly the wrong choice for long-range ranging in an absorbing liquid, and the usable band sits near and below 135 kHz. Panel 3 of Figure 3 shows the cliff directly; the curve is bounded to a readable range because past 250 kHz it descends through hundreds of negative dB.

5.2 A detection boundary near 135 kHz

With the reciprocity-consistent transducer and the modeled noise floor, the absolute SNRs at the three operating points are +14 dB at 135 kHz, +33 dB at 100 kHz, and -33 dB at 200 kHz, against a 12 dB detection threshold. The received-waveform panel (Figure 3, panel 4) tells the story in the time domain on per-frequency absolute scales: the 100 kHz echo is a clean 479 mV burst well above noise, the 135 kHz echo (57.6 mV) just clears the noise (the knife-edge case), and the 200 kHz echo (0.316 mV) is buried. Figure 4 shows the corresponding FDTD fields: at 100 kHz a wave packet survives the round trip and returns to the near end, while at 200 kHz it is absorbed in transit and never returns. The two views, a link budget and an independent wave solve, agree on the boundary.

5.3 The receiver, not the channel, is the limit

The modeled noise floor is dominated by the amplifier voltage noise times the large thin-film-PZT capacitance (Eq. (12)), giving an output-referred floor of order tens of millivolts. This is what determines detectability: the same channel against a naive $10 \mu\text{V}$ floor would appear roughly 70 dB more favorable. The levers that move the boundary are therefore on the receiver and the frequency, namely a lower transducer capacitance (thicker or bulk PZT, or charge matching at the element), a quieter front end, a narrower bandwidth or coded excitation, and above all a lower operating frequency to escape the absorption cliff. A range-by-frequency feasibility map (Figure 3, panel 6) shows the 12 dB contour crossing roughly 135 kHz at 3 m and opening up as either the range shortens or the frequency drops.

5.4 Why confinement matters

A free-field version of the same budget, target at 3 m in open oil, is dominated by spherical spreading and is at best marginal and strongly dependent on whether the target is a large specular reflector or a small scatterer. The tube removes that term: the wave is guided, the far end reflects nearly all of it, and the geometric penalty reduces to the aperture fill fraction. This is why a sealed oil-filled tube is a

far more forgiving channel than open water, and why the feasibility question turns on absorption and electronics rather than on spreading.

6 From detection to ranging precision

The application needs the piston position to about 0.1% of the 3 m stroke, roughly 3 mm. Low frequency appears to forfeit this. The classical axial resolution, about $c/(2B)$ with the resonant bandwidth $B \approx f_0/Q \approx 2.45$ kHz at 135 kHz, is near 29 cm. But axial resolution measures the ability to separate two closely spaced reflectors, which is not the task here.

The piston face is a single strong specular reflector ($\Gamma \approx 0.99$) at a slowly varying range. The relevant figure is the precision of the time-of-flight estimate to that one echo, bounded below by the Cramér–Rao limit, which tightens with SNR and is not capped by the pulse-envelope width; carrier-phase (I/Q) estimation reaches sub-wavelength precision at high SNR. For the round-trip phase $\varphi = 4\pi fd/c$ of a one-way distance d , the distance error is

$$\delta d \approx \frac{c}{4\pi f} \sigma_\varphi, \quad \sigma_\varphi \approx \frac{1}{\sqrt{2 \text{SNR}}}, \quad (13)$$

the second relation being the high-SNR phase bound. With the calibrated SNRs this conservative bound is $\sigma_d \approx 0.017$ mm at 100 kHz and 0.12 mm at 135 kHz. A matched-filter Monte-Carlo on the modeled echo plus modeled receiver noise (Appendix E), which integrates the ~ 10 -cycle burst coherently and so does at least as well, gives $\sigma_d \approx 3$ μm and 20 μm respectively. Both sit well inside the 3 mm target. The high $Q = 55$ helps rather than hurts, because the long coherent ring-down supplies many cycles over which to average phase. At 200 kHz and above the echo is below the 12 dB detection threshold, so there is no echo to time and ranging does not apply: the same f^2 cliff that forbids high frequency also defines the low band where phase ranging is precise.

Two practical points qualify this. The carrier phase wraps every half wavelength (about 5 mm one-way here), so an integer-cycle ambiguity must be resolved, by a coarse envelope or leading-edge estimate, or by tracking the slowly moving piston between pings. And absolute accuracy needs the oil sound speed $c(T, P)$ known to well under 0.1%, which a temperature sensor with a one-time calibration of the fixed electronic and wall-transit delays provides; phase methods are relatively insensitive to absolute amplitude calibration. Matched filtering against the channel-propagated template approaches the bound, and since receiver noise sets σ_φ , the receiver improvements that aid detection (lower e_n , lower C_s) tighten precision in the same proportion.

7 Limitations and failure modes

The dominant limitation is absolute calibration. The source paper’s transmit deflection sensitivity, its receive sensitivity from Eq. (8), and its tabulated design-map sensitivities disagree by more than $100\times$ when cross-checked, so any absolute SNR inherits roughly that uncertainty. I therefore present the absolute numbers as indicative and rest the conclusions on the relative structure, which is calibration independent: the f^2 cliff, the passivity constraint, the capacitance-limited noise regime, and the existence and approximate location of a detection boundary. The FDTD is a two-dimensional axial slice with a reduced steel speed and is used only for visualization. The oil attenuation and PZT permittivity are representative values, not measurements of a specific oil or device. There is no physical experiment; this is a modeling study whose purpose is to set design intuition and to provide a self-consistent, reproducible budget that a measurement campaign can later anchor.

8 Conclusion

For PMUT pulse-echo down a 3 m oil-filled steel tube, the conventional imaging design point near 1 MHz fails by a wide margin, because oil absorption over the round trip grows as f^2 and annihilates the echo. Confinement in the tube, not free-field radiation, makes the channel favorable, and once transmit and receive are tied to one coupling so the link is passive, the binding constraint is the receiver capacitance and amplifier noise rather than the acoustics. The usable band is near and below 135 kHz, and the design effort should go into lowering the operating frequency and the receiver noise floor. The model, the reciprocity relation, the noise budget, and the visualizations are reproducible from the released code.

9 Code and data availability

All code, figures, and the wave-propagation videos are released at <https://github.com/rayjyotir5/pmut-tube-echo>. The repository contains the 1D link-budget and noise model (`pmut_echo_sim.py`), the 2D FDTD solver and animation generator (`pmut_wave2d.py`), the figure scripts, the rendered 100/135/200 kHz movies, and a README with step-by-step reproduction instructions. The results figure and reports regenerate with `python3 pmut_echo_sim.py 100 135 200`; each FDTD movie with `python3 pmut_wave2d.py <freq_kHz>`.

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A Provenance: human input versus AI effort

This work was produced in a single interactive session in which a human supplied the starting paper, the physical scenario, a small set of physical parameters, and steering and quality feedback, while an AI agent (Claude, Anthropic, Opus 4.8) performed all reading, derivation, coding, debugging, figure generation, and writing. I log this explicitly so the division of labor is preserved.

A.1 Verbatim human prompts

Light typographic errors are preserved as written.

1. “read the modelling.pdf and understand what it is doing, will specify the actual problem after that”

2. “Has analytical model shown acoustic transmission through 10 mm steel + synthetic mineral oil yields detectable echo at 3 m with $\text{SNR} \geq 12$ dB at 1 MHz centre frequency? if not i want to set up a simulation and visualize the results” (clarifying choices: pulse-echo through wall; sweep noise; 1D analytical + transfer matrix)
3. “lets now upgrade the sim to a 2d environment, a steel tube is 3m long, 10mm thick and since it is the cross section of a circular tube, the outer dia of the tube is 300mm. the pmut is positioned at one end and the other end forms the reflecting surface (this end may have increased thickness of steel so as to ignore steel-air interface). tube is filled with oil. now run the simulation and repeat previous sweeps. feel free to re-center the frequency sweep and not assume 1MHz is a fixed input so that SNR is optimal. i’d also like to see a video or gif animation, use the WaveSimulator2D approach if required.”
4. “oil attenuation is 20 db/m and waveguide wall-bounce loss of 0.5db/m is fine. in the mp4, please render two more panels below the wave propagation animation. one is the pmut transmitter signal over time and a vertical line moves along synced to wave animation. similarly a panel for measured pmut receiver signal. also have a legend to show physical time of sim progress. hope the wave is simulated with noise considered at 250khz”
5. “erm 20db/m was not at 1mhz, it is at 250khz”
6. “render a second clip at the ~ 50 -100 kHz optimum so you can see a clean, surviving echo for contrast”
7. “is the sharp drop at 1MHz in the SNR vs frequency correct? why is this happening? would it help to now stop the sweep and model the noise correctly?”
8. “yes” (proceed to tie transmit and receive to one reciprocity-consistent coupling)
9. “lets now regenerate the results plots and wave propagation animations with the new calibrated model and view it above and below 135kHz”
10. “make sure to regenerate pmut_tube_results.png”
11. “i dont want three png files, i want a single one”
12. “the plots are at 250khz, pls update to 100, 135, 200”
13. “plot y axes are not scaled correctly, getting cropped”
14. “panel 1 and 4 as well, will log scale help?”
15. “i dont like the bar chart in panel 4, id rather see the waveform, not normalized to noise”
16. “treat this entire session as the research done, document everything and produce the resultant paper”

A.2 Session flow

Turn 1 established what the source model does. Turn 2 established that it does not address the ranging question and produced a first free-field link budget (two self-found bugs fixed: double-counted spreading, and an unphysical multi-watt transmit). Turn 3 reformulated the channel as a tube waveguide and added the 2D FDTD (a sign-error instability was found and fixed). Turns 4 to 6 set parameters, built the synchronized three-panel animation, corrected the attenuation reference frequency (a 112 dB change at the operating point), and rendered a low-frequency contrast clip. Turn 7 diagnosed the SNR cliff as the f^2 absorption law and replaced the assumed noise floor with a modeled one. Turn 8 tied transmit and receive together by reciprocity, removing an impossible gain-above-unity artifact. Turns 9 to 15 regenerated all outputs with the calibrated model and refined figure quality. Turn 16 produced this paper.

B Full parameter set

All values are documented in the released configuration (`src/pmut_echo_sim.py`).

Group	Quantity	Value
Geometry	tube length L	3.0 m
	outer diameter / wall	300 mm / 10 mm
	inner oil radius R_{in}	140 mm
	effective aperture radius a_{ap}	5 mm
Steel	density / speed	7850 kg/m ³ / 5900 m/s
	impedance	46.3 MRayl
Oil	density / speed	850 kg/m ³ / 1440 m/s
	impedance	1.22 MRayl
	absorption α_0 at 250 kHz	20 dB/m, $\propto f^2$
	waveguide wall-bounce loss	0.5 dB/m
Transducer	far-end reflection Γ	0.99
	quality factor Q	55
	$\Lambda', \Lambda_1, \Lambda_2$	0.99, 0.33, 0.20
	$e_{31,f}$	-24.24 C/m ²
	flexural rigidity D_e (calibrated)	1.0×10^{-5} N m
	plate eigenvalue α_{00}	3.19
	deflection sens. anchor	3 nm/V at 500 μm , 126 kHz
Receiver	feedback cap C_f	10 pF
	amplifier voltage noise e_n	4 nV/ $\sqrt{\text{Hz}}$
	PZT permittivity / thickness	1000 / 650 nm
	electrode fill / stray cap	0.7 / 5 pF
	dielectric loss $\tan \delta$	0.02
Operating	drive voltage V_{in}	10 V
	detection threshold	12 dB

Table 1: Model parameters. Steel speed is reduced to 2500 m/s in the FDTD only (density kept), to relax the time step while preserving the impedance contrast.

C Derivations

Wall transfer matrix. The intensity transmission of a single steel layer of impedance Z and thickness d between a source medium Z_1 and the oil Z_2 is

$$\tau_{\text{wall}} = \frac{4Z_1Z_2}{(Z_1 + Z_2)^2 \cos^2(kd) + (Z + Z_1Z_2/Z)^2 \sin^2(kd)}. \quad (14)$$

For a transducer well coupled to steel ($Z_1 = Z$) this reduces to the bare interface value $4Z_1Z_2/(Z_1 + Z_2)^2 \approx -10$ dB, independent of frequency; for an air-backed transducer ($Z_1 = Z_{\text{air}}$) the $\sin^2$ term creates deep nulls between the half-wave resonances at $n c_{\text{steel}}/2d$.

Confinement (fill) factor. In free space the on-axis far-field pressure of a piston of radius a falls as $\pi a^2/(\lambda z)$, so the monostatic round trip costs that factor squared, about -41 dB at $z = 3$ m. In the tube the launched power instead fills the cross-section and is conserved (apart from absorption); the only geometric loss is the fraction the receive aperture re-samples on return, $(a_{\text{ap}}/R_{\text{in}})^2$, which tends to unity as the aperture fills the tube.

Reciprocity relation. In the lumped circuit the transformer (ratio ϕ) maps the electrical port to the mechano-acoustic port with $P = \phi V$ and $I = \phi \dot{q}$. Transmit gives $\dot{q} = \phi V_{\text{in}}/Z_m$; receive gives $Q = \phi P_{\text{in}}/(j\omega Z_m)$. Eliminating ϕ and Z_m yields Eq. (11). Anchoring $\dot{q}/V$ to the measured 3 nm/V deflection and deriving Q/P_{in} from it makes the two paths share one coupling, so the two-way transfer cannot exceed what passivity allows.

Noise. The output-referred voltage-noise gain of a charge amplifier reading a source capacitance C_s is C_{in}/C_f with $C_{\text{in}} = C_s + C_{\text{stray}} + C_f$. The three terms in Eq. (12) are amplifier voltage noise, amplifier current noise through the feedback transimpedance $Z_f = 1/(\omega C_f)$, and the PZT dielectric-loss Johnson noise with parallel conductance $G = \omega C_s \tan \delta$, all integrated over ENBW = $(\pi/2)(f_0/Q)$.

FDTD update. The variable-density acoustic field uses a staggered velocity-pressure leapfrog: $\mathbf{v} \leftarrow (\Delta t/\rho)\nabla p$, then $p \leftarrow \Delta t \rho c^2 \nabla \cdot \mathbf{v}$, stable for $c_{\max}\Delta t/\Delta x \leq 1/\sqrt{2}$ in 2D. Open boundaries use a graded sponge layer; oil absorption is applied as a per-step decay $\exp(-\alpha c \Delta t)$ matched to Eq. (10) at the operating frequency. A sign error in the velocity update (using $+\nabla p$) made the scheme blow up; correcting it to $-\nabla p$ restored stability.

D Additional results

f (kHz)	oil loss (dB, RT)	echo	noise floor	SNR (dB)	$V_{\text{echo}}/V_{\text{in}}$
75	11	–	–	+46	0.168
100	19	479 mV	10.2 mV	+33.4	0.048
135	35	57.6 mV	11.9 mV	+13.7	0.0058
200	77	0.316 mV	14.5 mV	–33.2	3.2×10^{-5}
250	120	1.7 μ V	16.2 mV	–79	1.8×10^{-7}
1000	1920	~ 0	32 mV	–1898	4×10^{-98}

Table 2: Calibrated link at the design aperture (5 mm) and 3 m range. The detection boundary (12 dB) sits near 135 kHz. $V_{\text{echo}}/V_{\text{in}} < 1$ at every frequency confirms the model is passive after the reciprocity fix; before the fix it reached 14 at 75 kHz.

E Ranging precision (Monte-Carlo)

The estimator (`src/ranging_precision.py`) synthesizes the modeled echo (a Hann-windowed ~ 10 -cycle tone burst of amplitude V_{echo} at the operating frequency) plus modeled receiver noise of rms V_{noise} , and estimates the carrier phase over the known burst window with the maximum-likelihood single-bin estimator $\hat{\varphi} = \angle \sum_n r_n w_n e^{-j2\pi f t_n}$ (w_n the envelope). The circular standard deviation of $\hat{\varphi}$ over 2×10^4 noise realizations gives σ_{φ} , converted to a one-way distance error by Eq. (13). Coherent integration over the burst gives the matched-filter result a processing gain over the single-look bound computed from the narrowband link SNR, so the Monte-Carlo σ_d is smaller than the conservative CRLB figure.

f (kHz)	SNR (dB)	λ_{oil} (mm)	σ_d CRLB (mm)	σ_d MC (mm)
75	+45.5	19.2	0.006	0.001
100	+33.4	14.4	0.017	0.003
135	+13.7	10.7	0.124	0.020
200	–33.2	7.2	below 12 dB detection: no echo to time	
250	–79.3	5.8	below 12 dB detection: no echo to time	

Table 3: Single-reflector ranging precision. At the detectable low frequencies the one-way distance error is far below the 3 mm (0.1% of 3 m) target. At 200 kHz and above the echo is undetectable, so phase ranging does not apply; the phase “spread” there is the uniform wrap of an uninformative estimate, not a usable precision. Classical axial resolution at 135 kHz, by contrast, is $c/(2B) \approx 29$ cm with $B = f_0/Q$, but that is the two-reflector-separation figure, not the single-echo ranging precision.

F Calibration inconsistency

The source paper reports three sensitivity figures that do not agree when cross-checked. Its charge-sensitivity formula (Eq. (8)) evaluates to about 1130 fC/Pa for the configured stack, while the value implied by its measured 3 nm/V transmit deflection through the reciprocity relation is about 27 fC/Pa, a factor of 41. Separately, its design-map deflection sensitivity (Table 3, $\sim 0.42 \mu\text{m/V}$) and its measured deflection (Fig. 9, 3 nm/V) differ by more than $100\times$. I therefore treat absolute SNR as uncertain to roughly two orders of magnitude and rest the conclusions on the calibration-independent structure: the f^2 cliff, passivity, the capacitance-limited noise regime, and the existence of a detection boundary. Under any self-consistent choice within this spread, 250 kHz and above fail and the usable band is at low frequency.

G Reproducibility

All code, the rendered wave-propagation videos, the figures, and this paper are released at <https://github.com/rayjyotir5/pmut-tube-echo>. The repository README mirrors the steps below; after cloning, the commands reproduce every figure and movie.

```
git clone https://github.com/rayjyotir5/pmut-tube-echo
cd pmut-tube-echo/src
python3 pmut_echo_sim.py 100 135 200 # -> ../figures/pmut_tube_results.png + reports
python3 pmut_wave2d.py 100           # -> 100khz.mp4/.gif (also 135, 200)
python3 make_paper_figures.py        # -> schematic + wave montage
python3 ranging_precision.py         # -> ranging-precision sigma_d(f) table
cd ../paper
pdflatex main && bibtex main && pdflatex main && pdflatex main
```

Environment: Python 3.11, NumPy 2.0, SciPy 1.14, Matplotlib 3.10, imageio with the ffmpeg plugin; TeX Live 2025. The 1D link budget runs in seconds; each FDTD clip is a few minutes on a single CPU core.

H AI involvement checklist (summary)

Research stage assessment: ideation and framing were human-seeded (the scenario and the ranging question) and AI-developed; the literature anchor (one paper) was human-provided; experiment design, implementation, analysis, and writing were AI-led with human steering and feedback (High AI involvement). AI systems used: Claude (Anthropic), Opus 4.8, as an agent with code execution and figure generation. Known limitations and failure modes are stated in the main text (Limitations) and Appendix F: absolute calibration uncertainty, a visualization-only 2D solver, representative rather than measured material properties, and the absence of a physical experiment.